

CLARION, IA, USA

WMO: 725458

Lat: 42.742N

Lon: 93.759W

Elev: 354

StdP: 97.14

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04907

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.8	-20.3	-27.4	0.3	-22.9	-24.7	0.4	-20.9	14.8	-7.0	13.8	-8.6	3.5	320	0.601

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	32.7	24.0	32.0	23.6	30.0	22.8	26.5	30.5	25.4	29.4	24.2	28.5	5.8	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	21.2	29.2	23.9	19.7	28.2	22.7	18.2	26.8	84.6	30.6	80.0	29.4	74.9	28.6	31.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.8	11.5	10.4	-28.1	35.7	2.6	1.9	-30.0	37.1	-31.5	38.2	-33.0	39.3	-34.9	40.7		
(5)			-28.4	28.5	3.0	1.3	-30.6	29.4	-32.3	30.2	-34.0	30.9	-36.1	31.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	-7.5	-5.5	1.9	9.2	15.8	21.3	23.0	21.6	17.7	10.2	2.2	-5.2		
	DBStd	12.16	6.93	7.24	7.03	5.65	5.06	3.47	3.23	3.19	4.83	5.43	6.00	6.83		
	HDD10.0	2111	541	433	264	80	10	0	0	0	3	65	242	472		
	HDD18.3	3929	799	667	509	276	112	12	3	8	69	259	484	729		
	CDD10.0	1672	0	0	14	57	188	338	402	358	233	70	8	2		
	CDD18.3	448	0	0	1	3	32	101	147	108	50	6	0	1		
	CDH23.3	4292	0	0	9	68	347	934	1419	934	501	67	2	12		
	CDH26.7	1446	0	0	1	17	113	321	519	290	162	16	0	7		
(14) Wind	WSAvg	4.7	5.2	5.2	5.4	5.9	5.6	4.4	3.2	2.8	3.6	4.6	5.1	4.9		
(15) Precipitation	PrecAvg	906	23	25	52	93	124	150	124	111	72	57	45	29		
	PrecMax	1228	44	43	114	183	238	270	262	311	146	171	113	63		
	PrecMin	664	5	5	3	23	49	52	59	25	30	9	1	2		
	PrecStd	147	10	12	30	43	49	61	55	76	30	39	31	16		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.8	13.8	24.1	28.8	32.7	34.0	35.0	32.9	32.9	29.1	22.2	15.9		
		MCWB	5.8	11.2	15.7	16.4	19.8	23.7	26.3	24.6	22.9	19.3	14.5	12.7		
	2%	DB	4.9	9.4	19.0	25.1	29.8	32.3	32.7	32.0	30.9	25.0	17.4	8.6		
		MCWB	3.2	7.7	13.5	14.9	18.4	22.5	25.6	24.7	22.6	16.7	11.9	6.6		
	5%	DB	2.5	6.0	15.9	22.2	27.3	30.2	31.3	29.9	27.9	22.3	13.7	5.1		
		MCWB	1.6	4.0	11.8	13.3	17.6	21.7	24.7	24.2	20.3	15.9	9.3	3.2		
	10%	DB	1.3	2.9	12.1	18.9	24.8	28.0	29.8	27.8	26.3	19.1	11.0	2.7		
		MCWB	0.5	1.9	9.0	12.2	16.6	20.6	24.1	22.7	18.9	14.0	7.7	1.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.9	11.3	17.2	18.5	23.7	26.5	28.7	27.4	25.0	21.9	15.4	10.1		
		MCDB	8.6	13.5	23.3	24.9	28.1	30.7	33.2	31.0	29.7	26.1	20.1	12.5		
	2%	WB	3.2	7.6	14.3	16.4	21.3	25.0	26.8	26.1	23.5	19.1	12.4	5.8		
		MCDB	4.4	8.9	17.6	22.5	26.2	29.6	30.8	30.2	28.4	22.8	16.5	7.8		
	5%	WB	1.8	4.0	11.8	14.8	19.7	23.6	25.7	24.8	22.3	16.8	9.9	3.2		
		MCDB	2.5	5.5	16.2	20.5	24.5	28.2	29.6	28.6	26.7	20.2	13.0	4.8		
	10%	WB	0.4	2.2	8.6	12.9	18.1	22.3	24.6	23.6	20.9	14.7	7.6	1.6		
		MCDB	1.2	3.2	11.5	17.7	22.4	26.8	28.8	27.2	24.3	18.7	10.2	2.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	9.0	9.6	12.0	11.9	11.2	11.3	11.4	13.3	12.0	10.2	8.5		
		MCDBR	9.6	10.4	15.3	17.7	15.2	13.3	12.4	12.5	14.8	15.7	15.1	11.2		
	5% WB	MCWBR	8.4	8.7	9.8	9.4	6.9	6.5	7.2	7.5	7.4	8.7	9.6	8.9		
		MCWBR	9.4	10.1	14.3	14.5	12.1	10.9	11.0	10.8	12.3	12.5	13.6	10.1		
(40) Clear-Sky Solar Irradiance	taub	taub	0.279	0.291	0.317	0.370	0.398	0.422	0.428	0.412	0.384	0.338	0.296	0.274		
		taud	2.415	2.391	2.427	2.317	2.304	2.284	2.239	2.284	2.325	2.446	2.503	2.454		
	Ebn at Noon	Ebn at Noon	867	914	929	897	876	853	842	845	843	847	844	844		
		Edh at Noon	76	92	101	122	128	131	135	125	112	87	70	67		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.85	2.88	3.78	4.72	5.57	6.18	6.38	5.43	4.43	2.94	1.98	1.47		
	RadStd	RadStd	0.20	0.24	0.38	0.44	0.41	0.50	0.35	0.37	0.30	0.38	0.15	0.11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page